

Machine learning modeling of XGBoost and SVR for predicting rice husk dust ignition energy based on crushed brown rice content

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Highlights

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A trace amount of debris would reduce the ignition energy.

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Crushed brown rice can improve dispersibility, leading to a decrease in MIE.

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Hyperparameter optimization improves the prediction accuracy of the models.

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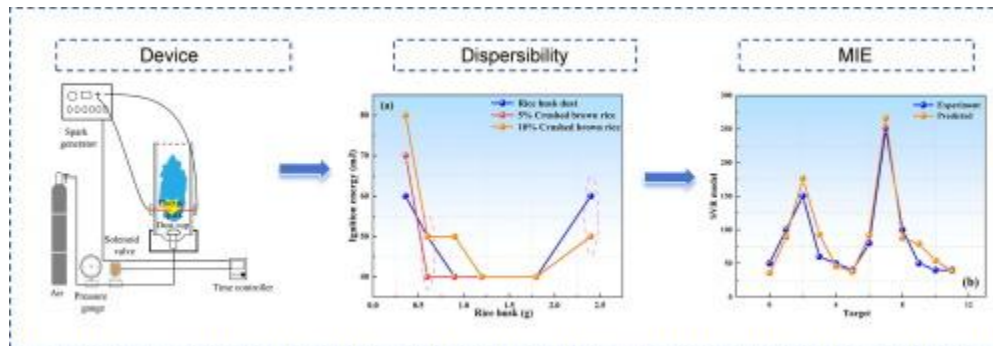
The SVR model is more suitable for predicting MIE.

Abstract

There exists the risk of dust explosion during the rice husk dust production and transportation. It is of great significance to clarify the parameter of the minimum ignition energy (MIE) of the dust cloud for reducing the dust explosion. This study explored the influence of the inclusion proportion of crushed brown rice (abbreviated as crushed brown rice) on the MIE of rice husk dust cloud. It was found that with the incorporation of crushed brown rice, the MIE of rice husk dust showed an increasing trend. However, a small amount of crushed brown rice can improve the dispersion of dust clouds, thereby reducing their ignition energy and increasing the risk of dust explosions. Moreover, two machine learning modeling methods, namely the Extreme Gradient Boosting algorithm (XGBoost) and the Support Vector Regression algorithm (SVR), were utilized to model the MIE of the rice husk.

The grid search was employed to automatically optimize the hyperparameters. In the optimal state of the model, the R^2 value of the SVR model, which is 0.913, is higher than that of the XGBoost model, which is 0.883, and the prediction error is lower at the same time, thus demonstrating that the SVR model is more suitable for the study of the MIE of the rice husk. Grain dust processing and transportation should therefore take these findings into consideration.

Graphical abstract



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Introduction

During the rice husk dust production and transportation, there exists a risk of dust explosion, and dust explosion accidents caused by rice husk powder occur from time to time [[1], [2], [3], [4]]. The MIE of the dust cloud is characterized as the energy that must be provided by the ignition source to ignite the dust, and MIE is an important indicator for evaluating the flammability of the dust [[5], [6], [7], [8]]. A better understanding of MIE is essential for preventing and mitigating the risks of fires and explosions in the process industries.

A vast number of existing studies have demonstrated that factors such as particle size, dust cloud concentration, dispersion pressure, ignition delay time, and moisture content make a big difference on the MIE of the dust cloud [[8], [9], [10], [11], [12]]. Measuring MIE through a large number of experiments is irreplaceable. However, dust explosion experiments are a statistical phenomenon, and the experimental results may vary with changes in environmental factors, equipment errors, and testing procedures. Moreover, high expenses are one of the difficulties associated with performing a lot of MIE tests, time consuming efforts, and safety concerns [[13], [14], [15]]. Predicting MIE is a crucial exercise that can save testing time, expenditure, and resources. Therefore, the development of an

accurate MIE prediction model will possess substantial usefulness in ensuring the safety of industrial processes.

Traditional computational theoretical models have been applied to the predictive research on MIE. However, due to the high degree of idealization and simplification of the research theoretical models, there still exists a deviation between the absolute experimental data and the calculated MIE data [[16], [17], [18], [19]]. In the last few years, new avenues for predictive modeling in the realm of dust explosion have been made possible by the proposal of several data-driven modeling techniques based on machine learning [[20], [21], [22]]. Machine learning algorithms, namely Random Forest (RF) and Artificial Neural Network (ANN), were utilized to develop models for predicting the MIE of 60 flammable compounds. It was verified that the RF algorithm is a promising option for MIE prediction applications [23,24]. Kevadiya, The MIE of 55 combustible organic compounds was successfully predicted using the Ordinary Least Squares (OLS), Huber, and Kernel Ridge Regression (KRR) models, group contribution, and machine learning techniques [5]. Moreover, the Back Propagation Neural Network (BPNN) prediction model and the Generalized Regression Neural Network (GRNN) prediction model were also applied to the forecast of the MIE of sugar powder clouds [25]. The aforementioned various models possess considerable predictive value in the field of MIE prediction. However, the focus of this study is on the MIE of different crushed brown rice contents based on machine learning modeling. The MIE of this type of mixed dust is affected by complex factors such as particle dispersion and interface interactions, and existing models have not been validated for their applicability in such scenarios. Therefore, the index of crushed brown rice content is selected as the input parameter. This is a novel experimental method, and it remains unknown whether the above models are applicable to the forecast of the MIE of dust clouds under such conditions. It is still difficult to choose the best machine learning technique for MIE prediction. The focus of our study is to compare different models for machine learning in order to accurately predict the MIE of rice husk dust clouds.

In this work, two models, XGBoost and SVR, were employed to predict the MIE of rice husk dust clouds under the condition of crushed brown rice admixture. By comparing the interpretability and prediction accuracy among the proposed machine learning models, a way to accurately predict the MIE of rice husk dust clouds under such conditions was sought. The significance of this work resides in the useful predictive tool it offers for the investigation of rice husk dust explosions in the industrial grain transportation process, enabling people to find methods to mitigate the damage caused by such accidents in the process industry.

Materials and devices

Rice husk dust and crushed brown rice are commonly found in the grain processing and transportation industries and thus are preferentially selected as experimental materials. The D_{50} of the rice husk powder samples is $17.56\ \mu\text{m}$ (Fig. 1a). The samples of large particles of crushed brown rice were acquired from a certain grain processing business in Liaoning (Fig. 1b). More introductions about our dust samples can be found in previous studies [1].

The MIE of the experimental samples was measured

Xgboost

XGBoost belongs to the integrated decision tree model (Fig. 3). The main idea of this algorithm is to continuously add trees and grow a tree by transforming features [22,29]. A new function will be learned to fit the residual of the prior prediction each time a tree is added. K trees are obtained at the end of the training process. Each tree will have a matching leaf node based on the sample attributes, and each leaf node is associated with a score. Ultimately, the sum of the scores yields the

Influence of large particles on the ignition energy of rice husk dust cloud

By adding samples of broken brown rice, the influence of mixing crushed brown rice in different ratios on the dust ignition energy was investigated. As the mass of rice husk dust increases, the dust ignition energy invariably exhibits a phenomenon of decreasing first and then increasing, and this phenomenon exists under any proportion of large particle admixture.

With the increase of the admixture proportion of large particles, the MIE of rice husk dust cloud shows an upward trend (Fig. 7).

Conclusion

In this study, the influence of the content of crushed brown rice on MIE was explored by mixing different proportions of large particles of crushed brown rice. We found that the mixing of large particles of crushed brown rice at a low proportion (5 %) caused a reduction in MIE, deviating from the expected trend. The crushed brown rice would improve the dispersibility of the mixed dust cloud, thus leading to the decrease in the ignition energy. Such crushed brown rice substances in rice

CRedit authorship contribution statement

Jinglin Zhang: Writing – review & editing. **Haiyan Chen:** Funding acquisition. **Zhenguo Du:** Conceptualization. **Shikai Bao:** Data curation. **Chang Li:** Investigation. **Gang**

Li: Investigation. **Caijun Bai:** Supervision. **Weitong Liang:** Project administration. **Zhiqun Xie:** Software. **Chunmiao Yuan:** Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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